

Sequential Eyelid Gestures for User Interfaces in VR

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ABSTRACT

In this poster, we present user-defined sequential eyelid gestures to control UIs in VR. Previous works have proposed sequential eyelid gestures for VR interaction. However, they do not include squint or wide-open eyelid states. In contrast, we consider those eyelid states too, and we encouraged users to prioritize how well the gestures fitted the commands. To validate that the user-defined gestures are effective, we tested them in a user study ($N = 17$) with five different UI commands for VR interfaces. We use the Vive Pro Eye HMD to detect all our final gestures with low error rates.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI; Interaction technique—Gestural input.

1 INTRODUCTION

The use of virtual reality (VR) has been steadily growing over the past few years. In part, the broader adoption of VR is thanks to all the efforts in academia to make it more accessible [3]. In this poster, we propose the design of sequential eyelid gestures for the most popular commands to control user interfaces (UI) in VR.

Motivated by the need to provide alternative hands-free input methods in VR, we designed our sequential eyelid gestures to perform the most standard commands needed to interact with UI in VR. These commands are push button, drag&drop, sliders, scale up, zoom in, scale down, zoom out, and change menu. Since the potential end-users in our interviews considered that the tasks of scaling and zooming are similar, we grouped them. We did the same with the tasks drag&drop and sliders. Therefore, we have 5 general commands to interact with UI in VR.

Most eye-based interaction research has focused on combinations with face or head gestures [5]. The idea of including gaze gestures is to reduce the Midas touch problem [4], where users perform unintentional command activations. In contrast, we use sequential eye gestures to overcome the Midas touch problem.

In general, our sequential eyelid gesture consists of 4 stages. For the initial stage, users' eyelids should be in a neutral position. The second stage starts when users perform an eyelid gesture and finishes when they recover a neutral eyelid state (stage 3). Finally, users perform a second eyelid gesture to trigger a command (stage 4). In Fig. 1, we present an overview of our sequential eyelid gestures, the commands that users can perform with them, and some action examples that those commands can trigger in VR interaction.

Previous works have designed sequential gestures with positive results; users find these gestures easy to use and remember. To the best of our knowledge, the work by Mingming et al. [2] is the only one to design sequential eyelid gestures. The main differences in

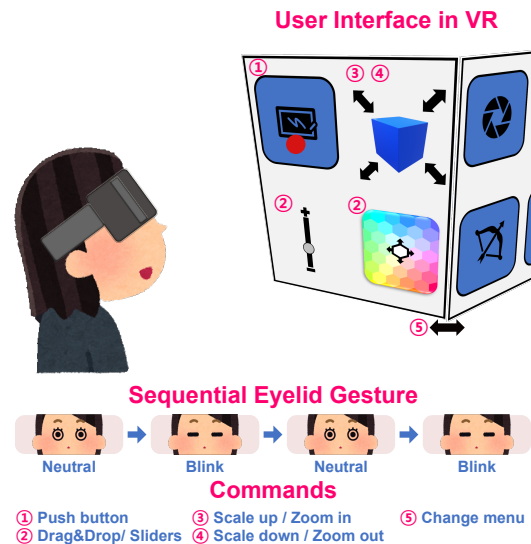


Figure 1: Sequential eyelid gestures for user interfaces in VR.

our approach are that we follow a user-defined design strategy, and we use a wider set of eyelid gestures (blink, blink the left eye, blink the right eye, squint, and wide-open eyes). By including a more diverse set of gestures, users can find variants that better fit the given command for VR interaction (e.g., using a combination that includes wide-open eyes to scale up a 3D model).

User-defined gestures have been successful in other contexts [1]. We follow a similar user-defined methodology, so we encouraged potential users of eyelid commands in VR to design the sequential eyelid gestures they want to perform.

Finally, we conducted a user study to validate that the user-defined gestures are effective. We tested them in a user study ($N = 16$) with five tasks (each task mimics a command). We designed the five tasks in VR to resemble the general use they have in the most popular VR apps in the market. Also, we use these tasks to measure the accuracy and execution error rate of the top three most-voted gestures for each command. To recognize the eyelid gestures, we use the dedicated hardware and software that the VIVE Pro Eye has for that purpose and a pair of myoelectric sensors to detect the wide-open eyes gesture more robustly.

2 METHOD

We have the objective to create user-defined sequential eyelid gestures for user interfaces (UI) in virtual reality (VR). Our method follows three steps to achieve this goal.

In step 1 (SP1), we gather user-defined *core eyelid gestures* ($N = 30$). These gestures are eyelid input modalities that users find easy and useful to perform for VR. Our selected gestures are blinking, one eye blinking, squinting, and wide-open eyes.

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Command	Agreement Score	Conceptual Difficulty	Top 3 Gestures		
			N1	N2	N3
Push Button	0.578	$\bar{x} = 2.667$ $SD = 0.942$	  → blink	 → blink	 → blink
Drag&Drop / Sliders	0.195	$\bar{x} = 6$ $SD = 0.816$	 → blink right eye	 → blink left eye	 → squint eyes
Scale up / Zoom in	0.352	$\bar{x} = 3.667$ $SD = 0.942$	 → wide open eyes	 → wide-open eyes	 → blink right eye
Scale down / Zoom out	0.258	$\bar{x} = 4.667$ $SD = 1.247$	 → squint eyes	 → squint eyes	 → blink left eye
Changing Menu	0.140	$\bar{x} = 3.333$ $SD = 1.24$	 → blink left eye	 → blink	 → blink right eye

Figure 2: The agreement scores for all the commands and their respective top 3 proposed gestures.

We perform a survey to elicit the most popular UI interactions for VR in step 2 (SP2). That is, we determine all the needed commands for which we need to map eyelid gestures. We found 5 actions for VR interaction: push button, drag & drop / sliders, scale up / zoom in, scale down / zoom out, and change menu.

In step 3 (SP3), we interview people in person ($N = 16$) to ask them which sequential eyelid gestures better fit each command we found in SP2. In Fig. 2, we present the agreement scores for all the commands and the top 3 most selected gestures. As an introduction, we made sure to explain to participants that the most relevant qualities of the selected gestures should be how well they fit the command and how easy they are to perform. Besides, we mentioned to participants that they had to provide a different gesture for each command.

Finally, in SP3, we found that users have a higher agreement score on commands such as push button, scale up / zoom in, and scale down / zoom out. On the other hand, users do not agree much on assigning gestures to the commands drag&drop and change menu.

3 USER STUDY

Users found it difficult to agree on what gesture to assign for each command (see Fig. 2). To resolve the conflicts on our user-defined gestures, we include a user study in which participants test the top three gestures for each command. Therefore, we have a user study with 5 tasks (one per command). Broadly speaking, we designed the tasks so users can test the gestures in scenarios that provide a general view of the command they represent.

3.1 The Participants

Students from our university volunteered to participate in our user study ($N = 17$, 5 female). All of them have previous experience using VR and good eye health (1 left-handed). Their age range is from 21 and 36 years ($\bar{x} = 23.06$, $SD = 3.386$), and 47.06% of them have used eye-tracking devices.

3.2 The Procedure

Once participants tested all gestures in a task, we asked them to order the gestures – according to how well they fit the task – from best to worse. Finally, we did an exploratory interview to understand their selection criteria. The completion of a task and its set of gestures takes around 10 minutes. The total duration of the user study is ~ 1.5 hours.

4 DETERMINATION OF USER-DEFINED SEQUENTIAL EYELID GESTURES

To follow our user-defined strategy, we propose to resolve gesture conflict by replacing gestures with the lower agreement score with their next best. For example, since the agreement score for the best gesture *Task 1 - target selection* (0.433) is lower than the counterpart of *Task 5 - iteration* (0.446), we update it to its second-best gesture (blink left eye → blink). Likewise, we assign to *Task 2 - drag&drop* the gesture blink → blink right eye.

For *Task 1 - target selection*, we assigned blink left eye → blink. This gesture got the second spot in the raking and aligns with the qualitative and quantitative results; it is the second best gesture in *learning*, *comfort*, and *execution time*.

In *Task 2 - drag&drop* the assigned gesture is blink → squint (third place in the ranking). This gesture is the slowest in the ranking.

The selected gesture for scaling up objects (*Task 3*) is blink → blink right eye (first place in the ranking). The selected gesture is the easiest to perform.

We assigned the first place gesture in the raking (blink → blink left eye) to *Task 4 - scale down*, which is the easiest along with the second place in the ranking.

For *Task 5 - iteration*, we selected blink → blink (first place in the ranking). The selected gesture is one of the easiest to learn and the fastest to perform.

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